

Entropy is an

- ☐ a. Additive property
- ☐ b. Partially intensive & partially extensive
- ☐ c. Intensive property
- ☐ d. Extensive property

The natures of lattice and basis in a respective order are:

- ☐ a. Both are real
- ☐ b. Real and Imaginary
- ☐ c. Imaginary and Real
- ☐ d. Both are Imaginary

An 820-N Marine in basic training climbs a 12.0-m vertical rope at a constant speed

- ☐ a. 4.23 KW
- ☐ b. 1.23 KW
- ☐ c. 3.23 KW
- ☐ d. 2.23 KW

An astronaut hits a golf ball on the Moon. Which of the following quantities, if any

- ☐ a. Horizontal component of velocity
- ☐ b. Speed
- ☐ c. Velocity
- ☐ d. Vertical component of velocity

In the limit of high temperature, the particles are distributed over a large range of energy v

- ☐ a. Very large
- ☐ b. Zero
- ☐ c. Very small
- ☐ d. Same

Force due to magnetic field and velocity is

- ☐ a. At acute angles with each other
- ☐ b. Antiparallel to each other
- ☐ c. At obtuse angle with each other
- ☐ d. At right angles to each other

The intensity of the black body radiation is proportional to the wavelength (λ^{-4}) :

- ☐ a. Wien's displacement law
- ☐ b. Stefan's law
- ☐ c. Rayleigh-jeans law
- ☐ d. Planck's law

The intensity of the black body radiation is proportional to the wavelength (λ^{-4})

- ☐ a. Wien's displacement law
- ☐ b. Stefan's law
- ☐ c. Rayleigh-jeans law
- ☐ d. Planck's law

Which of the following is false?

- ☐ a. Electric susceptibility of the medium is dimensionless
- ☐ b. Permittivity of the material is not dimensionless
- ☐ c. The dielectric constant of the material is not dimensionless
- ☐ d. All

When nuclear radiations through gas ionization is produced, this is the principle of wh

- ☐ a. Geiger Muller counter
- ☐ b. Proportional counter
- ☐ c. Flow counter
- ☐ d. Scintillation counter

A parallel-plate capacitor is filled with insulating material of dielectric constant ϵ_r . What Electric field?

- ☐ a. Increasing
- ☐ b. Decreasing
- ☐ c. No effect
- ☐ d. None of the above

Which of the following is true about an n-type material?

- ☐ a. The hole is the majority carrier and the electron is the minority carrier
- ☐ b. Both the electron and the hole are majority carriers
- ☐ c. It is formed using a trivalent impurity
- ☐ d. The electron is the majority carrier and the hole is the minority carrier

A capacitor stores 0.24 coulombs at 10 volts. Its capacitance is

- ☐ a. 0.12 F
- ☐ b. 0.024 F
- ☒ c. 0.6 F
- ☐ d. 0.8 F

Clear my choice

An isotope with a high Binding Energy per nucleon:

- ☐ a. Has very few electrons
- ☐ b. Will decay in a short period of time
- ☐ c. Is very unstable
- ☐ d. Is very stable

ation

	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
--	---	---	---	---	---	---	----	----	----	----	----	----	----	----	----

X-ray diffraction occurs when:

- ☐ a. The incoming rays are totally reflected
- ☒ b. The scattered rays interfere constructively
- ☐ c. The incoming rays are totally absorbed by the crystal
- ☐ d. The scattered rays interfere destructively

Clear my choice

ack

tion 56

et
red
d out of
question

s page

avigation

Time left 1.3

One of the following is not the postulate of quantum mechanics for a particle in one dimension?

- ☒ a. To each state of a physical system there corresponds a wave function $\Psi(x, t)$
- ☐ b. The time development of the wave function is determined by the Schrodinger equation
- ☐ c. To every physical observable b there corresponds a Hermitian operator B
- ☐ d. The answer is not given

One of the following is not the postulate of quantum mechanics for a particle in one dimension

- ☐ a. To each state of a physical system there corresponds a wave function $\Psi(x, t)$
- ☐ b. The time development of the wave function is determined by the Schrödinger equation
- ☐ c. To every physical observable b there corresponds a Hermitian operator B^*
- ☐ d. The answer is not given

The splitting of a nucleus into smaller nuclei is

- ☐ a. Half-life
- ☒ b. Fission
- ☐ c. gamma -radiation
- ☐ d. Fusion

What is conserved in the first law of thermodynamics?

- ☐ a. Energy
- ☐ b. Work
- ☐ c. Mass
- ☐ d. Heat

Energy band gap of an insulating material is:

- ☐ a. Less than 5 eV
- ☐ b. Greater than 5 eV
- ☐ c. Equal to 1 eV
- ☐ d. 0 eV

Why are nuclear energy levels more complex than electron energy levels?

- ☐ a. Nuclear energy levels are an order of one hundred times as great as electron energy levels
- ☐ b. Electron energy levels depend on the interaction between neutrons and electrons
- ☐ c. Nuclear energy levels depend on attractive and repulsive forces
- ☐ d. Nuclear energy levels depend only on attractive forces

If a gas is compressed isothermally, which of the following statements is true?

- ☐ a. Energy is transferred into the gas by heat
- ☐ b. The temperature of the gas increases
- ☐ c. The internal energy of the gas remains constant
- ☐ d. No work is done on the gas

The dot product of the vectors $A=2\hat{i}-3\hat{j}+\hat{k}$ and $B=4\hat{i}+3\hat{j}+\hat{k}$ is

- ☐ a. -1
- ☐ b. 2
- ☐ c. 1
- ☐ d. 0

The purpose of doping is to:

- ☐ a. Give a semiconductor material certain properties.
- ☐ b. Make the charge carriers move faster.
- ☐ c. Protect devices from damage in case of transients
- ☐ d. Cause holes to flow.



ion

If z be the partition function of a system of particles, the average energy is written as

- ☐ a. $\bar{E} = -\frac{\partial z}{\partial \beta}$
- ☐ b. $\bar{E} = \frac{\partial \ln z}{\partial \beta}$
- ☐ c. $\bar{E} = \frac{\partial z}{\partial \beta}$
- ☐ d. $\bar{E} = -\frac{\partial \ln z}{\partial \beta}$

The direction of electric field due positive charge is

- ☐ a. Away from the charge
- ☐ b. Towards the charge
- ☐ c. Both A and B
- ☐ d. None of the above

The dimensions of generalized momentum

- ☐ a. May be those of the linear momentum
- ☐ b. May be those of angular momentum
- ☐ c. Are always those of linear momentum
- ☐ d. A and B

Work done by external force in N-particle system is known as ____

- ☐ a. Work
- ☐ b. Virtual work
- ☐ c. Total work
- ☐ d. None of these

Work done by external force in N-particle system is known as ____

- ☐ a. Work
- ☐ b. Virtual work
- ☐ c. Total work
- ☐ d. None of these

The rate of transfer of energy by an electromagnetic wave is equal to

- ☐ a. $\frac{EB}{\mu_0}$
- ☐ b. $\frac{E^2}{\mu_0 c}$
- ☐ c. $\frac{cB^2}{\mu_0}$
- ☐ d. All of the above

A collection of microstates of a system all having one or more extensive parameters in common is called

- ☐ a. Statistical ensemble
- ☐ b. Thermodynamic system
- ☐ c. Set of microstates
- ☐ d. A macrostate